TALIBON POLYTECHNIC COLLEGE

BED 106 — BUSINESS ANALYTICS

Mini Business Analytics Capstone Project

Academic Year 2026–2027 • 1st Semester • Instructor: Jessie A. Melendres

Course Code	BED 106
Course Title	Business Analytics
Number of Units	3 Units — 54 Hours
Document Type	Semester Performance Task — Mini Capstone Project
Checkpoints	4 Checkpoints (Preliminary, Midterm, Semi-Final, Final)
Total Points	400 points (100 pts per checkpoint)
Mode	Group Project (2–4 members)
A.Y. / Semester	2026–2027 / 1st Semester

SECTION 1 — Project Overview

The Mini Business Analytics Capstone Project is a semester-long performance task that challenges students to act as junior data analysts solving a real-world business problem. From selecting a business domain in Week 1 to defending a fully integrated analytics solution in Week 18, students progressively build on each topic covered in the course.

The project is structured around four (4) checkpoints aligned with the four grading periods of the course. Each checkpoint assesses the skills and competencies introduced in that phase, while building toward a comprehensive final output.

1.1 Course Intended Learning Outcomes Addressed

This capstone project is designed to assess all four Course Intended Learning Outcomes (CILOs):

- CILO 1 — Apply data fundamentals, data quality concepts, and SQL to extract, organize, and prepare business data for analysis.
- CILO 2 — Apply spreadsheet and statistical techniques (descriptive statistics, correlation, regression) to analyze business data and support decision-making.
- CILO 3 — Design and build interactive dashboards and data visualizations using business intelligence tools, and differentiate components of BI/data warehousing architecture.
- CILO 4 — Apply predictive analytics concepts and evaluate ethical, privacy, and governance issues in business data use, integrating them into a defensible capstone analytics project.

1.2 What is the Project?

Each group will select ONE business domain and ONE specific business problem to investigate throughout the semester. Using real or realistic data, the group will:

1. Collect, clean, and explore a relevant business dataset using SQL.
2. Perform descriptive statistical analysis and regression using spreadsheet tools.
3. Build an interactive dashboard and visual story using Power BI or Tableau.
4. Apply predictive analytics, address ethical considerations, and present a final integrated report and defense.

1.3 Approved Business Domains

Groups must choose from the following business domains (or propose an alternative with instructor approval):

Retail & Sales Analytics	**Human Resources Analytics**	**Financial Analytics**
Sales trends, inventory turnover, customer segmentation, revenue forecasting	Employee attrition, performance vs. salary, headcount planning	Expense tracking, budget vs. actual, profit margin analysis
Healthcare Analytics	**Supply Chain Analytics**	**Education Analytics**
Patient trends, appointment analytics, readmission rates	Supplier performance, delivery lead time, demand forecasting	Student performance, enrollment trends, dropout prediction

1.4 Group Requirements & Individual Accountability

- Groups must be composed of 2 to 4 members, formed during Week 1.
- Every group member must be assigned a specific role (see Section 1.5).
- Individual contribution pages must be signed and submitted with every checkpoint.
- The instructor reserves the right to ask any member to explain any part of the project during checkpoints and the final defense.
- Peer evaluation (10% of Checkpoint 4) will be conducted to measure individual contribution.

1.5 Suggested Team Roles

Role	Primary Responsibilities
Project Lead / Analyst	Oversees timelines, coordinates submissions, leads business problem framing, and handles documentation.
Data Engineer	Responsible for data collection, cleaning, SQL query development, and database management.
Statistician / Modeler	Handles descriptive statistics, correlation, regression analysis, and predictive modeling tasks.
BI Developer / Visualizer	Builds the dashboard in Power BI or Tableau; designs charts, KPIs, and the final presentation deck.

1.6 Data Sources

Students are encouraged to use real-world datasets from the following sources:

- Kaggle (www.kaggle.com/datasets)
- Philippine Statistics Authority (psa.gov.ph)
- World Bank Open Data (data.worldbank.org)
- Local business data (with proper consent and anonymization)

- Instructor-provided sample datasets

All datasets must be documented with source, date accessed, and any license information. Datasets must contain at minimum 200 rows to allow meaningful analysis.

SECTION 2 — Checkpoint Overview

The project is divided into four (4) checkpoints aligned with the four grading periods. Each checkpoint builds on the previous one, ensuring a progressive and integrated capstone experience.

CP	Phase	Focus Area	Weeks	Pts
CP 1	Preliminary	Data Fundamentals & SQL Querying	Weeks 1 – 4	**100**
CP 2	Midterm	Spreadsheet & Statistical Analysis	Weeks 5 – 9	**100**
CP 3	Semi-Final	BI Dashboard Development	Weeks 10 – 13	**100**
CP 4	Final	Capstone Analytics Project & Defense	Weeks 14 – 18	**100**
TOTAL PROJECT SCORE				**400**

Each checkpoint submission is graded independently out of 100 points. The total capstone project score of 400 points is factored into the course Project grade (20% of final grade).

CHECKPOINT 1 — Data Fundamentals & SQL Querying

Preliminary Period • Weeks 1–4 • 100 Points

Objectives

- Define a business problem and scope the analytics project.
- Identify and document a relevant business dataset.
- Design and populate a relational database with clean, quality data.
- Write SQL queries to extract and summarize key business metrics.

Tasks & Deliverables

Task 1.1 — Business Problem Statement

Write a one-page problem statement that includes:

- The business domain and the specific problem you are investigating.
- Why this problem matters (business impact / significance).
- The key business questions you aim to answer through analytics.
- The type of data you plan to use and where you will source it.

Task 1.2 — Dataset Documentation

Provide a Data Source Report that includes:

- Dataset name, source URL or file, date accessed, and license type.
- A data dictionary: for each column, state the name, data type, and description.
- A preliminary data quality assessment: identify missing values, duplicates, inconsistencies, and what cleaning steps were applied.
- A screenshot or printed preview of the raw dataset (first 10–20 rows).

Task 1.3 — Relational Database Design

Using MySQL or any SQL-compatible RDBMS:

- Design an Entity-Relationship Diagram (ERD) with at least 2 related tables.
- Create the tables with appropriate data types and primary/foreign keys.
- Import and load the cleaned dataset into the database.
- Provide screenshots of the schema and populated tables.

Task 1.4 — SQL Query Report

Write and execute a minimum of eight (8) SQL queries demonstrating:

- At least 2 SELECT queries with WHERE and ORDER BY clauses.
- At least 2 GROUP BY queries with aggregate functions (COUNT, SUM, AVG, MAX, MIN).
- At least 2 JOIN queries connecting two or more tables.
- At least 2 business-relevant queries that directly answer your key business questions.

For each query, provide: (a) the SQL code, (b) the output/result set screenshot, and (c) a 2–3 sentence business interpretation.

Checkpoint 1 Rubric

Criterion	Points	Description / What to Submit
Problem & Data Understanding	**20 pts**	Business problem is clearly defined with specific questions. Dataset is well-documented with a complete data dictionary and quality assessment. Problem is scoped appropriately for the semester.
SQL Query Accuracy	**35 pts**	All 8+ SQL queries execute correctly without errors. Queries include SELECT, GROUP BY, JOIN, and aggregate functions. Queries are logically structured and relevant to the business problem.
Data Preparation	**25 pts**	ERD is properly designed with at least 2 related tables, appropriate keys, and data types. Database is populated with clean data. Data quality issues are identified and resolved.
Documentation	**20 pts**	Report is organized, complete, and professionally formatted. All screenshots are labeled. SQL code is readable with comments. Business interpretations are insightful and accurate.
TOTAL	**100 pts**	

Submission Requirements

- One (1) printed and bound project report (Tasks 1.1–1.4).
- SQL script file (.sql) submitted via class portal.
- Signed Individual Contribution Form (template provided by instructor).
- Submission deadline: End of Week 4 laboratory session.

CHECKPOINT 2 — Spreadsheet & Statistical Analysis

Midterm Period • Weeks 5–9 • 100 Points

Objectives

- Apply Excel formulas, pivot tables, and charts to analyze business data.
- Compute and interpret descriptive statistics for business variables.
- Perform and interpret a correlation analysis between key metrics.
- Build and interpret a simple linear regression model to support a business decision.
- Create a basic forecast using trend analysis.

Tasks & Deliverables

Task 2.1 — Spreadsheet Analytics Report

Using Microsoft Excel or Google Sheets, prepare a structured spreadsheet workbook with the following sheets:

- Sheet 1 — Cleaned Data: The finalized version of your dataset imported into the spreadsheet.
- Sheet 2 — Pivot Analysis: At least 3 pivot tables summarizing key business metrics (e.g., total sales by category, average by region, count by period).
- Sheet 3 — Pivot Charts: At least 3 charts derived from the pivot tables with proper titles, axis labels, and business annotations.
- Sheet 4 — Formulas Showcase: Demonstrate at least 5 Excel functions relevant to your business context (e.g., SUMIF, COUNTIF, VLOOKUP/XLOOKUP, IF, TEXT).

Task 2.2 — Descriptive Statistics Report

For at least 3 numerical variables in your dataset, compute and report:

- Measures of central tendency: Mean, Median, Mode.
- Measures of dispersion: Standard Deviation, Variance, Range, Min, Max.
- Frequency distribution table and histogram for one key variable.
- A written narrative (1 paragraph per variable) interpreting what these statistics reveal about the business.

Task 2.3 — Correlation Analysis

Identify two pairs of variables that may be related in your business context. For each pair:

- Compute the Pearson Correlation Coefficient using Excel's CORREL function or Data Analysis ToolPak.
- Create a scatter plot with a trendline.
- Interpret the direction, strength, and business implication of the correlation.

Task 2.4 — Simple Linear Regression

Select one dependent variable (Y) and one predictor variable (X) relevant to your business question. Perform a regression analysis using the Excel Data Analysis ToolPak and:

- State the regression equation (Y = a + bX).
- Report and interpret the R-squared value.
- Test the significance of the slope (p-value).
- Use the equation to make at least one business forecast or prediction.
- Discuss the limitations of the model and conditions for its use.

Task 2.5 — Trend / Seasonality Analysis (if applicable)

If your dataset contains time-series data, add a forecasting sheet that:

- Plots the data over time and identifies the trend pattern.
- Projects the next 3–6 periods using Excel's FORECAST or trendline function.
- Discusses the reliability and assumptions of the forecast.

Checkpoint 2 Rubric

Criterion	Points	Description / What to Submit
Spreadsheet Report Quality	**25 pts**	Excel workbook is well-organized with labeled sheets. Pivot tables and charts are correct, properly formatted, and relevant. Formulas are accurate and appropriate for the business context.
Statistical Accuracy	**30 pts**	Descriptive statistics are correctly computed and complete. Correlation coefficients and regression outputs are accurate. Calculations match the data and tool outputs are verified.
Interpretation of Results	**30 pts**	Written narratives show genuine business insight from each analysis. The regression model is interpreted correctly including R^2, p-value, and the regression equation. Findings are tied back to the original business questions.
Documentation	**15 pts**	Report is neatly structured with clear headings. Charts are titled and labeled. Spreadsheet file is submitted alongside the printed report. Analysis logic is explained clearly.
TOTAL	**100 pts**	

Submission Requirements

- Printed report containing all written interpretations and printed screenshots of outputs.
- Excel workbook (.xlsx) file submitted digitally via class portal.
- Updated Signed Individual Contribution Form.
- Groups must retain the same dataset used in Checkpoint 1.
- Submission deadline: End of Week 9 laboratory session.

CHECKPOINT 3 — BI Dashboard Development

Semi-Final Period • Weeks 10–13 • 100 Points

Objectives

- Apply principles of effective data visualization to design a clear and insightful dashboard.
- Build an interactive, multi-page dashboard using Power BI or Tableau.
- Apply basic data mining concepts (clustering or segmentation) to the dataset.
- Present the dashboard to the class and articulate the visual insights.

Tasks & Deliverables

Task 3.1 — Dashboard Design Blueprint

Before building, submit a hand-drawn or digital wireframe/mockup of the planned dashboard that includes:

- A layout sketch of each dashboard page/tab (at least 3 pages).
- The title and purpose of each page.
- The types of charts/visuals planned (bar, line, pie, map, KPI cards, etc.) and what each will show.
- Identification of at least 2 interactive elements (slicers, filters, drill-through).

Task 3.2 — Interactive BI Dashboard

Build the final dashboard using Power BI Desktop or Tableau Public. The dashboard must include:

- Page 1 — Executive Summary: Key Performance Indicators (KPIs) relevant to the business problem (at least 4 KPI cards).
- Page 2 — Trend & Comparison Analysis: Time-series or categorical comparison charts. At least 2 different chart types (e.g., line + bar).
- Page 3 — Deep Dive / Segmentation: Detailed visuals supporting your analytical story, with at least one visual from your clustering/segmentation task.
- Interactive slicers or filters that allow the user to slice data by at least 2 dimensions (e.g., period, region, category).
- All visuals must be properly titled, labeled, and sourced.

Task 3.3 — Data Mining: Clustering or Segmentation

Apply a basic clustering or segmentation analysis to your dataset:

- Use Excel K-Means, Python (if familiar), or the built-in clustering features in Power BI/Tableau.
- Identify at least 2–3 meaningful customer/product/employee segments.
- Label each segment with a descriptive business name (e.g., “High-Value Customers,” “Seasonal Buyers”).
- Visualize the segments in the dashboard and explain the business significance of each cluster.

Task 3.4 — Dashboard Presentation (In-Class)

Each group will present their dashboard live to the class during Weeks 12–13. The 10-minute presentation must:

- Walk through all three dashboard pages while narrating the business story.
- Demonstrate at least one interactive feature (filter/slicer) live.
- Answer at least 2 questions from the instructor or classmates.
- Every member must speak during the presentation.

Checkpoint 3 Rubric

Criterion	Points	Description / What to Submit
Dashboard Design & Usability	**30 pts**	Dashboard follows design principles: appropriate chart types, readable fonts, consistent colors, logical layout, and clear titles. Interactive elements work correctly. Navigation between pages is intuitive.
Data Accuracy	**25 pts**	All values and visualizations match the underlying dataset. KPIs are correctly calculated. Clustering/segmentation is logically applied and accurately represented.
Insights & Storytelling	**25 pts**	The dashboard tells a coherent business story. Each visual has a clear purpose. Insights are data-driven and directly address the business questions. The narrative flows from overview to detail.
Documentation	**20 pts**	Blueprint wireframe is submitted. Written report explains design decisions, key insights, and segmentation findings. Dashboard file is exported and submitted in the required format.
TOTAL	**100 pts**	

Submission Requirements

- Printed report: Blueprint wireframe + written discussion of insights and segmentation.
- Dashboard file: Power BI (.pbix) or Tableau (.twbx) submitted digitally.
- Exported PDF version of all dashboard pages.
- Updated Signed Individual Contribution Form.
- Live dashboard presentation scheduled during Week 12 or 13 (schedule assigned by instructor).
- Submission deadline: Beginning of Week 14.

CHECKPOINT 4 — Capstone Analytics Project & Final Defense

Final Period • Weeks 14–18 • 100 Points

Objectives

- Apply a predictive analytics concept to the business problem.
- Evaluate and articulate the ethical, privacy, and governance implications of the data and analysis.
- Integrate all prior checkpoints into a polished, professional capstone report.
- Defend the project in a formal presentation before the instructor and peers.

Tasks & Deliverables

Task 4.1 — Predictive Analytics Application

Building on the regression model from Checkpoint 2, extend your predictive analytics:

- Apply at least one predictive technique: Multiple Regression, Logistic Regression (conceptual), Decision Tree outline, or trend-based forecasting.
- Present the model, its variables (predictors and output), and its predictions.
- Evaluate model performance using R^2, accuracy rate, or another appropriate metric.
- Provide at least 3 business predictions or actionable recommendations based on the model output.
- Discuss the assumptions, limitations, and conditions of your predictive model.

Task 4.2 — Data Ethics, Privacy & Governance Report

Write a minimum 500-word reflection addressing the following:

- Privacy Considerations: What personal or sensitive data (if any) is in your dataset? How was it handled (anonymized, consented, licensed)?
- Ethical Issues: Are there any biases in the data or model? Could the analysis be used to unfairly discriminate against any group?
- Data Governance: Who owns the data? What policies govern its use? What would a responsible data governance framework look like for this dataset?
- Regulatory Awareness: Reference at least one relevant regulation or framework (e.g., Republic Act 10173 — Data Privacy Act of the Philippines, GDPR concept).

Task 4.3 — Integrated Capstone Report

Compile all four checkpoint outputs into one cohesive, professionally formatted final report with the following structure:

#	Report Section
1	Title Page — Project title, group members, roles, course info, date.

2	Executive Summary — 1-page synopsis of the business problem, methods used, and key findings.
3	Business Problem & Data Source — Updated and refined from Checkpoint 1.
4	Database Design & SQL Analysis — Final ERD, cleaned data documentation, and SQL query results.
5	Statistical Analysis — Descriptive statistics, correlation, and regression from Checkpoint 2.
6	BI Dashboard Summary — Screenshots and discussion of all dashboard pages from Checkpoint 3.
7	Predictive Analytics — Model description, predictions, and recommendations (Task 4.1).
8	Ethics & Governance — Full reflection report (Task 4.2).
9	Conclusions & Recommendations — Synthesis of all findings with 3–5 actionable business recommendations.
10	References — All datasets, tools, and literature cited in APA format.
11	Appendix — Individual Contribution Forms, raw data sample, peer evaluation forms.

Task 4.4 — Final Presentation & Defense

Each group will defend their capstone project in a 20–25 minute formal presentation during Weeks 16–18. The defense schedule will be assigned by the instructor.

Presentation Structure:

- Introduction & Business Context (2–3 min): Briefly reintroduce the business problem and team.
- Data & SQL Overview (3–4 min): Show the ERD and highlight 2–3 key SQL findings.
- Statistical Insights (3–4 min): Present the most important descriptive and regression findings.
- Live Dashboard Demo (5–6 min): Walk through the Power BI / Tableau dashboard live, demonstrating filters.
- Predictive Analytics & Recommendations (4–5 min): Present the predictive model and 3–5 actionable recommendations.
- Ethics & Governance (1–2 min): Summarize key ethical considerations.
- Open Defense / Q&A (5 min): Respond to questions from the instructor and panel.

Checkpoint 4 Rubric

Criterion	Points	Description / What to Submit
Business Problem & Scope	15 pts	The business problem is clearly defined and consistently addressed throughout all sections of the final report. Scope is appropriate and the project is well-grounded in a real business context.
Analysis & Predictive Technique	25 pts	Predictive model is correctly applied and interpreted. Model evaluation metrics are reported. Predictions are logical and supported by the data. Ethics and governance reflection demonstrates critical thinking.

Dashboard / Report Quality	**25 pts**	Final integrated report is complete, professionally formatted, and covers all 11 sections. Dashboard screenshots are clearly annotated. Visuals are polished and consistent with findings.
Presentation & Defense	**20 pts**	Presentation is clear, well-structured, and within the time limit. All members contribute. The team answers Q&A confidently and correctly. Visual aids (slides and dashboard) are effectively used.
Documentation & Peer Eval	**15 pts**	Individual Contribution Forms are signed and complete. Peer evaluation forms are submitted. References are properly cited in APA. Appendix is complete.
TOTAL	**100 pts**	

Submission Requirements

- Final Capstone Report (spiral-bound hardcopy + digital PDF).
- All digital files (SQL, Excel, Power BI/Tableau, datasets) compiled in one organized folder via class portal.
- Presentation slide deck (.pptx or .pdf).
- Completed Peer Evaluation Forms (one per member, signed).
- Signed Individual Contribution Forms for all four checkpoints.
- Report and digital files due: Beginning of Week 16 (before defense schedules begin).
- Defense dates: Weeks 16–18 (scheduled by instructor).

SECTION 3 — General Guidelines & Policies

3.1 Formatting Standards

- All written reports must use Arial 12pt font, 1-inch margins, and 1.5 line spacing.
- Section headings must be clearly labeled and consistently formatted.
- All figures, tables, and charts must be numbered and captioned.
- Cover page must include: course code and title, project title, group members and roles, submission date, and academic year.

3.2 Academic Integrity

- All analysis must be the group's own work. Plagiarism of text, data, or analysis from other groups or online sources is strictly prohibited.
- AI-generated analysis (e.g., using ChatGPT to write interpretations) is not permitted. Insights must be derived by the students themselves.
- Datasets from public sources must be properly cited; data fabrication is grounds for a failing mark.
- The instructor may request any group member to individually explain any part of the project at any time.

3.3 Late Submissions

Days Late	Penalty
1 day late	10% deduction from the checkpoint score.
2 days late	20% deduction from the checkpoint score.
3 days late	30% deduction from the checkpoint score.
More than 3 days	Checkpoint receives zero (0). No further submissions accepted.
Excused (with documentation)	Full credit possible if excused absence is documented and approved by the instructor.

3.4 Revision & Consultation Policy

- Groups are encouraged to consult with the instructor during lab sessions in Weeks 5, 9, 13, and 15 (designated consultation weeks).
- Groups may revise and resubmit Checkpoint 1 and Checkpoint 2 outputs with Checkpoint 3, but revised scores are capped at 80 out of 100.
- No revisions are accepted after the final defense in Week 18.

3.5 Tools & Software

Checkpoint	Primary Tool	Alternatives Accepted
CP 1 — SQL	phpMyAdmin	PostgreSQL, SQLite, MySQL Workbench
CP 2 — Spreadsheet	Microsoft Excel	Google Sheets (with Data Analysis add-on)
CP 3 — BI Dashboard	Power BI Desktop (Free)	Tableau Public (Free), Google Looker Studio
CP 4 — Predictive / Report	Excel + PowerPoint	Python (pandas, sklearn) with instructor approval

SECTION 4 — Master Project Schedule

Week	Phase	Key Project Milestone	Checkpoint
1	Preliminary	Form groups; choose domain & problem; begin dataset search	CP 1 Begins
2	Preliminary	Data collection & cleaning; design ERD; create SQL schema	
3	Preliminary	Write SQL queries; prepare Data Source Report	
4	Preliminary	Finalize CP 1 report and SQL script — SUBMISSION WEEK	**CP 1 Due**
5	Midterm	Import cleaned data to Excel; build pivot tables & charts	CP 2 Begins
6	Midterm	Compute descriptive statistics for key variables	
7	Midterm	Correlation analysis and scatter plots	
8	Midterm	Linear regression and trend/forecast analysis	
9	Midterm	Finalize CP 2 report and Excel workbook — SUBMISSION WEEK	**CP 2 Due**
10	Semi-Final	Design dashboard blueprint/wireframe; connect data to BI tool	CP 3 Begins
11	Semi-Final	Build Dashboard Pages 1 & 2 (KPIs, trends)	
12	Semi-Final	Build Dashboard Page 3; apply clustering; rehearse presentation	
13	Semi-Final	Dashboard presentations (live in class) — SUBMISSION WEEK	**CP 3 Due**
14	Final	Develop predictive analytics model; write ethics reflection	CP 4 Begins
15	Final	Integrate all sections into final capstone report	

16	Final	Submit complete report & digital files; defense begins	**CP 4 Submissions Due**
17	Final	Final defense presentations	
18	Final	Final defense presentations COURSE WRAP-UP	**CP 4 Due (Defense)**

SECTION 5 — Appendix Forms

Form A — Individual Contribution Form

To be completed and signed for every checkpoint submission.

Individual Contribution Form — BED 106 Capstone Project

Group Name / Number: ______________________________ Checkpoint No.: ______

Member Name: ______________________________ Role: ______________________________

Specific Contributions This Checkpoint:

1. ____
2. ____
3. ____

I certify that the contributions listed above are accurate and reflect my actual participation.

Signature: ________________________ Date: ________________

Form B — Peer Evaluation Form (Checkpoint 4 Only)

Each member individually rates ALL other group members. Submit to the instructor in a sealed envelope on defense day.

Peer Evaluation Form — Final Capstone Defense

Evaluator's Name: ______________________________ Group: ______________________

Rate each member below (1 = Poor, 5 = Excellent) and add comments:

Member Name	Effort	Quality	Collab.	Commun.	Comments
Member 1: ____________________	/5	/5	/5	/5	
Member 2: ____________________	/5	/5	/5	/5	
Member 3: ____________________	/5	/5	/5	/5	

Evaluator's Signature: ________________________ Date: ________________

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